Software Engineering Lab 7

Fuzz testing for input validation bugs

SS1 | Original application run

Before changes: the supplied application processes ordinary sample inputs.

```
Microsoft Windows [Version 10.0.26200.9711]
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C:\Users\ayush\OneDrive\Desktop\SEM5\SE_LAB\PES1UG24AM909_SWE_LAB\LAB7>cd /d "C:\Users\ayush\OneDrive\Desktop\SEM5\SE_LA
B\PES1UG24AM909_SWE_LAB\LAB7"

C:\Users\ayush\OneDrive\Desktop\SEM5\SE_LAB\PES1UG24AM909_SWE_LAB\LAB7>python restaurant_bill_parser.py
==== Running clean_dish_name ====
Enter a dish name from the menu (e.g. *paneer tikka*): *paneer tikka*
Dish name: Paneer Tikka

==== Running parse_item_prices ====
Enter item prices separated by commas (e.g. 250,180,320): 250,180,320
Item prices: [250, 180, 320]

==== Running split_amount ====
Enter the bill and number of people as TOTAL/PEOPLE (e.g. 1200/4): 1200/4
Each person pays: 300
```

Hypothesis findings

SS2 | Tests failing against the original application

```
C:\Users\ayush\OneDrive\Desktop\SEM5\SE_LAB\PES1UG24AM909_SWE_LAB\LAB7>python -m pytest -v test_restaurant_bill_parser.py
============================================= test session starts =============================================
platform win32 -- Python 3.11.9, pytest-9.1.1, pluggy-1.7.0 -- C:\Users\ayush\AppData\Local\Programs\Python\Python311\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\ayush\OneDrive\Desktop\SEM5\SE_LAB\PES1UG24AM909_SWE_LAB\LAB7
plugins: hypothesis-6.170.15
collected 3 items

test_restaurant_bill_parser.py::test_clean_dish_name_no_crash FAILED                                 [ 33%]
test_restaurant_bill_parser.py::test_parse_item_prices_no_crash FAILED                               [ 66%]
test_restaurant_bill_parser.py::test_split_amount_no_crash FAILED                                    [100%]

================================================== FAILURES ==================================================
_____________________________________ test_clean_dish_name_no_crash _____________________________________
```

Failure summary

```
=========================================== short test summary info ===========================================
FAILED test_restaurant_bill_parser.py::test_clean_dish_name_no_crash - Failed: clean_dish_name crashed for None: 'NoneType' object has no attribute 'strip'
FAILED test_restaurant_bill_parser.py::test_parse_item_prices_no_crash - BaseExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_restaurant_bill_parser.py::test_split_amount_no_crash - BaseExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
============================================ 3 failed in 3.80s ============================================
```

After the fixes

SS3 | Fixed application run with malformed price and split inputs

```
C:\Users\ayush\OneDrive\Desktop\SEM5\SE_LAB\PES1UG24AM909_SWE_LAB\LAB7>python restaurant_bill_parser.py
==== Running clean_dish_name ====
Enter a dish name from the menu (e.g. *paneer tikka*): *panner tikka*
Dish name: Panner Tikka

==== Running parse_item_prices ====
Enter item prices separated by commas (e.g. 250,180,320): 250,abc,-10
Item prices: [250]

==== Running split_amount ====
Enter the bill and number of people as TOTAL/PEOPLE (e.g. 1200/4): 1200/0
Each person pays: None
```

Final test results

SS4 | Hypothesis tests passing against the fixed application

```
C:\Users\ayush\OneDrive\Desktop\SEM5\SE_LAB\PES1UG24AM909_SWE_LAB\LAB7>python -m pytest -v test_restaurant_bill_parser.py
============================================================ test session starts ============================================================
platform win32 -- Python 3.11.9, pytest-9.1.1, pluggy-1.7.0 -- C:\Users\ayush\AppData\Local\Programs\Python\Python311\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\ayush\OneDrive\Desktop\SEM5\SE_LAB\PES1UG24AM909_SWE_LAB\LAB7
plugins: hypothesis-6.170.15
collected 3 items

test_restaurant_bill_parser.py::test_clean_dish_name_no_crash PASSED                                                                  [ 33%]
test_restaurant_bill_parser.py::test_parse_item_prices_no_crash PASSED                                                                [ 66%]
test_restaurant_bill_parser.py::test_split_amount_no_crash PASSED                                                                     [100%]

============================================================ 3 passed in 1.92s =============================================================

C:\Users\ayush\OneDrive\Desktop\SEM5\SE_LAB\PES1UG24AM909_SWE_LAB\LAB7>
```